

Unit-3

⇒ Laser:-

- Einstein's Theory of laser action,
- Einstein's coefficients population inversion and lasing action.
- Construction and working of He-Ne and semiconductor lasers.
- Properties of laser beam.
- Applications of laser in science and engineering.

⇒ Optical Fiber:-

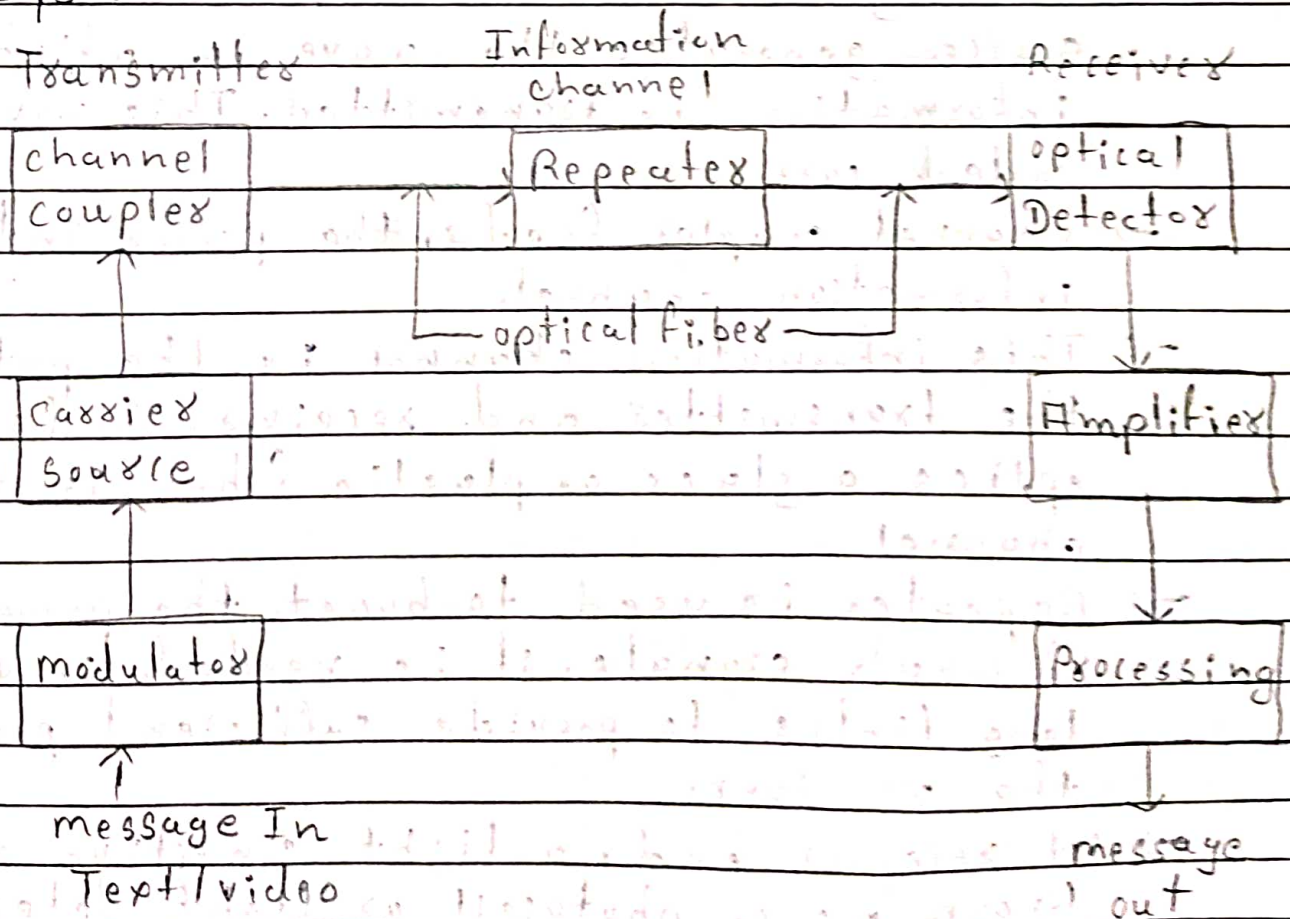
- structure
- Types
- Features
- Light guiding mechanism
- Acceptance angle and Numerical aperture.

Unit-3

Optical fibre:-

- An optical fibre is a thin flexible transparent fibre that acts as a wave guide or light pipe to transmit light betⁿ the two end of the fibre.
- Optical fibre are widely used in optical fibre communication which permits transmission over longer distances and higher bandwidths than other form of communication.
- Fibres are used instead of metal wire because signals travel along them with less loss and are also immune to electromagnetic interference.

⇒ Block Diagram of optical fibre communication system.



→ Basic Block Diagram of optical fiber communication system:

→ The basic optical fiber communication system consists of following parts.

1. Transmitter
2. Information channel
3. Receiver

⇒ Block diagram of OFC system:

⇒ Generally message origin is from a transducer that converts a non-electrical signal into electrical signals.

→ ex:- Microphones.

⇒ Modulator impresses this signal onto the wave generated by the carrier source. Carrier source generates the wave on which the information is transmitted. This wave is called carrier.

→ Channel couples feeds the power into information channel.

→ This information channel is the path between the transmitter and receiver. In fiber optics a glass or plastic fiber is the channel.

→ Repeater is used to boost the power level of weak signals. It is needed in very long links to provide sufficient power to the receiver.

→ At receiver end, a light sensitive device known as a photocell or light detector is used to detect the information. This photo

detector converts the light signal back to electrical signal. The electrical signals are amplified and reshaped back into digital form.

⇒ Structure of optical fiber:-

→ Optical fiber cable consists of four parts

1. Core
2. Cladding
3. Buffer
4. Jacket

1. Core:-

→ The core of a fiber cable is a cylinder of plastic that runs all along the fiber cable's length, and offers protection by cladding. The diameter of the core depends on the application used. Due to internal reflection, the light travelling within the core reflects from the core, the cladding boundary. The core cross-section needs to be a circular one for most of the applications.

2. Cladding:-

→ Cladding is an outer optical material that protects the core. The main function of the cladding is that it reflects the light back into the core. When light enters through the core into the cladding, it changes its angle, and then reflects back to the core.

3. Buffer:-

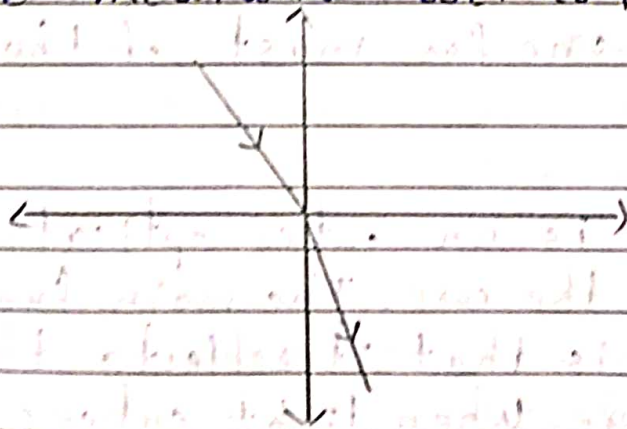
→ The main function of the buffer is to protect the fiber from damage and thousands of optical fibers arranged in hundreds of optical cables. These bundles are protected by the cable's outer covering that is called jacket.

4. Jacket:-

→ Fiber optic cable's jackets are available in different colors that can easily make us recognize the exact color of the cable we are dealing with. The color yellow clearly signifies a single mode cable, and orange color indicates multimode.

Refraction of light:-

→ As a light ray passes from one transparent medium to another it changes direction this phenomena is called refraction of light depend on the medium whether it is rarer or denser.



→ How much the light ray changes its direction depends on the refractive index of the medium.

Refractive Index:-

→ Refractive index is the speed of light in vacuum divided by the speed of light in a material. Refractive index measures how much the light refracts. It is abbreviated as $n = \frac{c}{v}$

$\therefore c = \text{speed of light in vacuum}$
 $v = \text{speed of light in medium}$

Snell's law:-

→ When light passes from one transparent material to another bends according to Snell's law.

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

where $n_1 = \text{refractive index of } m-1$

$n_2 = \text{refractive index of } m-2$

$\theta_1 = \text{incident angle}$

$\theta_2 = \text{refractive angle}$

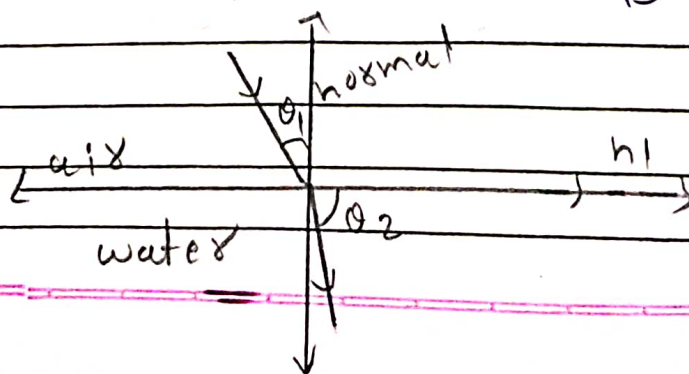
Critical angle:-

→ The critical angle can be calculated from Snell's law putting in the angle of 90° for the angle of the refracted ray for $\theta_2 = 90^\circ$

Since $\theta_2 = 90^\circ$

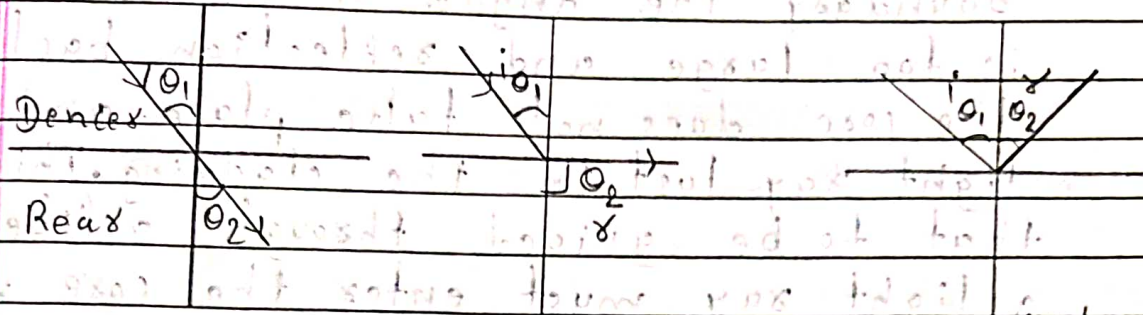
$$\sin \theta_2 = 1$$

$$\theta_c = \theta_1 = \arcsin\left(\frac{n_2}{n_1}\right)$$



Principal of optical fiber:-

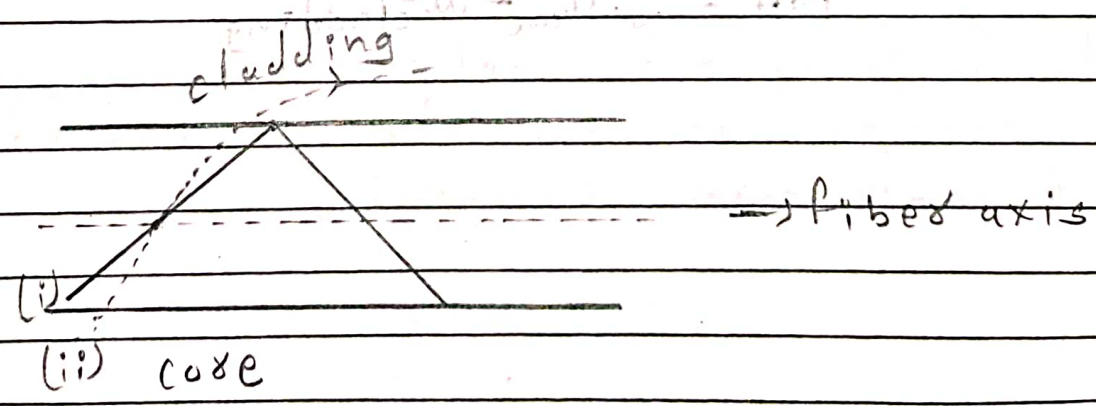
→ Total internal reflection:-



$\theta_2 = 90^\circ$ For total internal reflection.
critical angle

→ If the light hits the interface at any angle larger than any critical angle. It will not pass through the second medium at all. Instead all of it will be reflected back into the first medium. This process is known as total internal reflection.

Numerical Aperture and Max. angle of Acceptance:-

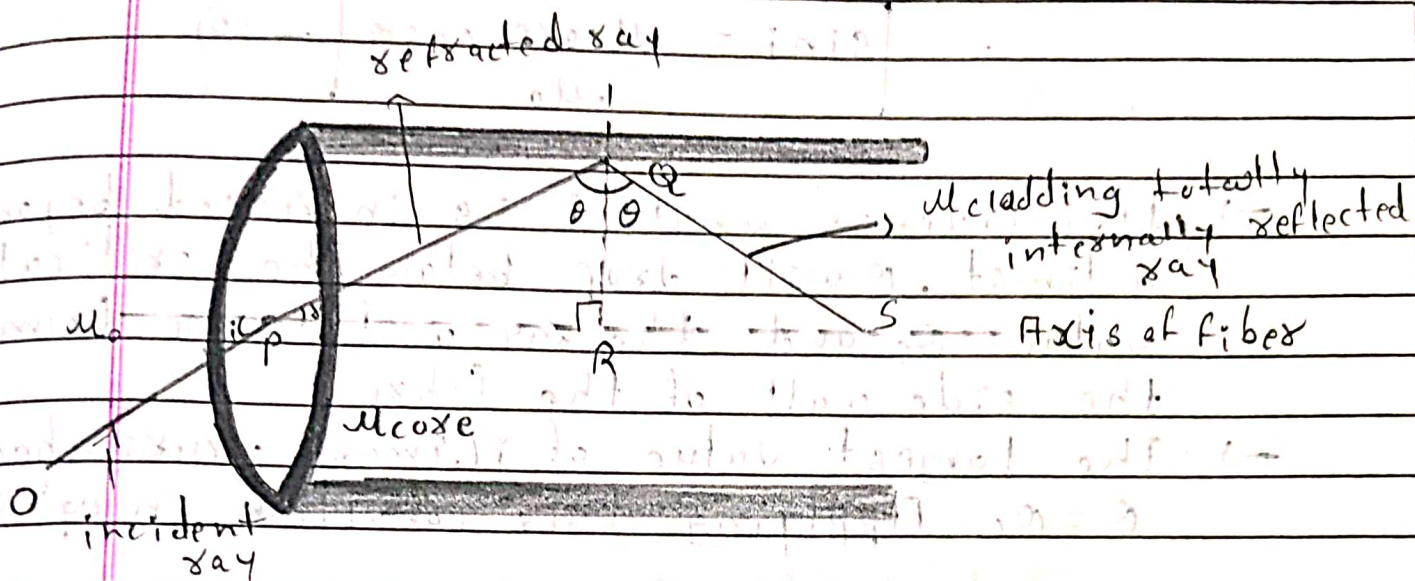


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- If a ray enters the fiber at a steep angle for example light ray (ii) then when this ray intersects the core/cladding boundary the angle of intersection is too large and reflection back into the core does not take place, and the light ray is lost in the cladding. This means that to be guided through a fiber a light ray must enter the core with an angle that is less than the so called acceptance angle for the fiber.
 - By convention the acceptance angle for a fiber can also be described in terms of numerical aperture.
 - The numerical aperture of a fiber is simply equal to the mathematical sign of the fiber acceptance angle. Hence the numerical aperture is given by:

$$\theta_A = \sin^{-1} \sqrt{\mu_{\text{core}}^2 - \mu_{\text{cladding}}^2}$$

$$NA = \sqrt{\mu_{\text{core}}^2 - \mu_{\text{cladding}}^2}$$

#) Derivation of Numerical Aperture:-



⇒ $\mu_o \rightarrow$ Refractive index of medium where light ray enters the fiber

$\mu_{core} \rightarrow$ R.I of core

$\mu_{clad} \rightarrow$ R.I of cladding

$i \rightarrow$ incident angle

$r \rightarrow$ refracted angle

$\theta \rightarrow$ Angle at core-clad interface

$\theta_c \rightarrow$ Critical angle.

→ Applying Snell's law at a point of entry of the ray OP into the core we have

$$\mu_o \sin i = \mu_{core} \sin r \quad \text{--- (1)}$$

From ΔPQR $r + \theta = 90^\circ$

$$r = 90^\circ - \theta$$

Taking sin on both side

$$\sin r = \sin (90^\circ - \theta)$$

$$\sin r = \cos \theta \quad \text{--- (2)}$$

substituting the value of $\sin r$ in eqn-①

$$\begin{aligned} \mu_0 \sin i &= \mu_{\text{core}} \cos \theta \\ \boxed{\sin i &= \frac{\mu_{\text{core}} \cos \theta}{\mu_0}} \quad \text{--- ③} \end{aligned}$$

- If incidence angle i is increased beyond a limit, θ will drop below the critical value θ_c and the ray will escape from the side wall of the fibre.
- The largest value of i (i_{max}) occurs when $\theta = \theta_c$. Applying this condition in eqn-③.

$$\sin(i_{\text{max}}) = \left(\frac{\mu_{\text{core}}}{\mu_0} \right) \cos \theta_c \quad \text{--- ④}$$

→ At core-cladding surface

$$\mu_{\text{core}} \sin \theta_c = \mu_{\text{clad}} \sin 90^\circ$$

$$\sin \theta_c = \frac{\mu_{\text{clad}}}{\mu_{\text{core}}}$$

$$\boxed{\sin \theta_c = \frac{\mu_{\text{clad}}}{\mu_{\text{core}}}}$$

$$\begin{aligned} \rightarrow \cos \theta_c &= \sqrt{1 - \sin^2 \theta_c} \\ &= \sqrt{1 - \left(\frac{\mu_{\text{clad}}}{\mu_{\text{core}}} \right)^2} \end{aligned}$$

$$= \frac{\mu^2 \cos \theta_c - \mu_{clad}^2}{\mu^2 \cos \theta_c}$$

→ Put the value of $\cos \theta_c$ in eqn (4)

$$\sin(i_{max}) = \frac{\mu \cos \theta_c}{\mu_0} \sqrt{\frac{\mu^2 \cos \theta_c - \mu_{clad}^2}{\mu^2 \cos \theta_c}}$$

$$= \sqrt{\frac{\mu^2 \cos \theta_c - \mu_{clad}^2}{\mu^2 \cos \theta_c}} \times \frac{\mu^2 \cos \theta_c}{\mu_0^2}$$

$$= \sqrt{\frac{\mu^2 \cos \theta_c - \mu_{clad}^2}{\mu_0^2}} \quad \text{For air } \mu_0 = 1$$

$$\sin(i_{max}) = \sqrt{\mu^2 \cos \theta_c - \mu_{clad}^2}$$

$$i_0 = \sin^{-1} \sqrt{\mu^2 \cos \theta_c - \mu_{clad}^2}$$

i_0 → acceptance angle

N.A is $\sin$ of i_0

$$N.A = \sin \left[\sin^{-1} \left(\sqrt{\mu^2 \cos \theta_c - \mu_{clad}^2} \right) \right]$$

$$N.A = \sqrt{\mu^2 \cos \theta_c - \mu_{clad}^2}$$

Propagation conditions:-

→ If i is the angle of incidence of an incident ray then the ray will be able to propagate only if i is i_0 .

where $i_0 = \sin^{-1} \sqrt{\mu_{\text{core}}^2 - \mu_{\text{clad}}^2}$

or

$$\sin i < \sin i_0$$

or

$$\sin i < \sqrt{\mu_{\text{core}}^2 - \mu_{\text{clad}}^2}$$

This is called propagation condition.

V-number:-

→ This is an important parameter of optical fiber given by the relation.

$$V = \frac{2\pi a}{\lambda} \sqrt{\mu_{\text{core}}^2 - \mu_{\text{clad}}^2}$$

where $a \rightarrow$ radius of core

$\lambda \rightarrow$ free space wavelength

→ The max. no. of modes (N_m) supported by a single step index fiber is determined by

$$N_m \approx \frac{1}{2} V^2$$

→ If $V < 2.405$, the fiber will support only one mode and known as single mode fiber.

→ If $V > 2.405$, the fiber will support many mode simultaneously known as multi mode fiber.

→ Wavelength corresponding to the value 2.405 known as cut off wavelength

$$\lambda_c = \frac{\lambda V}{2.405}$$

⇒ Fractional refractive index change (Δ):-

→ The fractional refractive index change is defined as the ratio of refractive index difference betⁿ the core and cladding to the refractive index of the core of fiber.

$$\Delta = \frac{\mu_{core} - \mu_{clad}}{\mu_{core}}$$

N.A. in terms of (Δ)

$$N.A. = \sqrt{\mu_{core}^2 - \mu_{clad}^2}$$

$$N.A. = \sqrt{(\mu_{core} + \mu_{clad})(\mu_{core} - \mu_{clad})}$$

$$N.A. = \frac{(\mu_{core} + \mu_{clad})(\mu_{core} - \mu_{clad})}{2\mu_{core}} \cdot 2\mu_{core}$$

Now $\frac{\mu_{core} + \mu_{clad}}{2} \approx \mu_{core}$

$$\mu_{core} \approx \mu_{clad}$$

$$N.A. = \int \mu_{core} \cdot \Delta \cdot 2 \mu_{core}$$

$$= \int 2 \Delta \mu_{core}^2$$

$$N.A. = \mu_{core} \sqrt{2\Delta}$$

① calculate the refractive index of a core and cladding materials of fiber given that $N.A. = 0.22$, $\Delta = 0.012$.

Solⁿ:-

$$N.A. = \mu_{core} \sqrt{2\Delta}$$

$$\frac{N.A. 0.22}{\sqrt{2 \times 0.012}} = \mu_{core}$$

$$\mu_{core} = \frac{0.22}{\frac{0.15}{\sqrt{0.024}}} = \frac{0.22}{0.15} = 1.46$$

$$\boxed{\mu_{core} = 1.46}$$

$$\rightarrow \Delta = 1 - \frac{\mu_{clad}}{\mu_{core}}$$

$$0.012 = 1 - \frac{\mu_{clad}}{1.46}$$

$$0.012 - 1 = - \frac{\mu_{clad}}{1.46}$$

$$= -0.988$$

$$-0.988 \times 1.46 = -\mu_{clad}$$

$$1.44 = \mu_{clad}$$

$$\boxed{\mu_{clad} = 1.44}$$

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Q. Show + Calculate the numerical aperture and acceptance angle, C.A, having $\mu_{\text{core}} = 1.50$, $\mu_{\text{clad}} = 1.45$.

Soln:-

$$N.A = \sqrt{\mu_{\text{core}}^2 - \mu_{\text{clad}}^2}$$

$$= \sqrt{(1.50)^2 - (1.45)^2}$$

$$= \sqrt{2.25 - 2.10}$$

$$= \sqrt{0.15}$$

$$N.A = 0.38$$

$$\rightarrow \theta_A = \sin^{-1} N.A$$

$$= \sin^{-1}(0.38)$$

$$\theta_A = 22.33$$

$$\rightarrow \theta_c = \sin^{-1} \left(\frac{1.45}{1.50} \right)$$

$$= \sin^{-1}(0.96)$$

$$\theta_c = 73.73$$

- ③ Calculate the maximum radius allowed for a fiber having $n_{core} = 1.47$ and $n_{clad} = 1.46$. Fiber has a support of only one mode at wave length of 1300 nm . What is the

$$V = \frac{2\pi a}{\lambda} \sqrt{n_{core}^2 - n_{clad}^2}$$

- ④ An optical fiber has $NA = 0.20$, $n_{clad} = 1.59$. Determine the OA for the fiber in water which has refractive index 1.33 .

$$88.0 = NA$$

$$NA = 0.20 = NA$$

$$(88.0) \cdot NA =$$

$$88.88 = NA$$

$$(88.0) \cdot NA = NA$$

$$(88.0) \cdot NA =$$

$$88.88 = NA$$

Classification of fiber:-

(1) According to modes

(A) Single mode fiber

(B) Multimode fiber

(2) According to refractive index

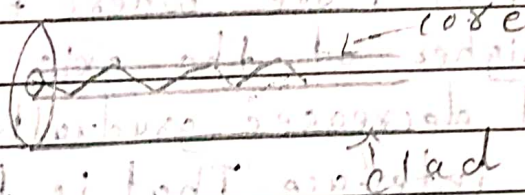
(A) Step index fiber

(B) Graded index fiber

(A) Single mode fiber:-

→ In a fiber if only one mode is transmitted through it then it is said to be a single mode fiber.

→ The core radius of single mode fiber is nearly equal 3 micrometer and numerical aperture is equal to 0.1 at a wavelength of 0.8 micrometer and V -number is < 2.405 .

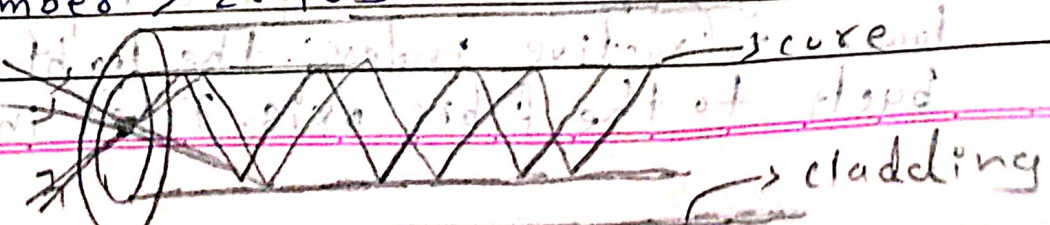


(B) Multimode fiber:-

→ If more than one mode is transmitted through optical fiber is called multimode fiber.

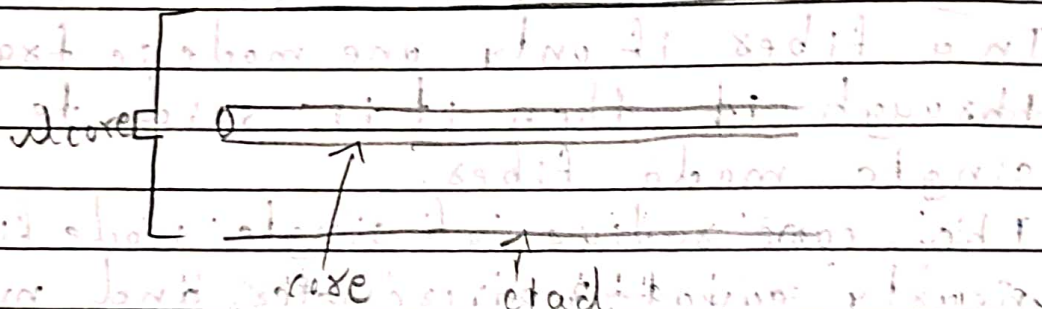
→ Multimode fiber has larger core radius make it easier to launch optical power to fiber.

→ V number > 2.405 .



(A) Step-index fiber:-

- In step index fiber the refractive index changes in step fashion from the centre of the fibre the core to the outer shell to the cladding.
- In these fiber (mode) the light in the fiber propagate by bouncing back and forth from core cladding interface.
- Step-index fiber propagate both single and multimode signals.



(B) Graded-Index Fiber:-

- Graded Index multimode fiber is a type of optical fiber where the refractive index is higher at the axis of the core and then it decreases gradually towards the core cladding interface. That is to say the refractive index of a graded index fiber gradually decreases from a centre and eventually decreases to the same value as the cladding at the core edge.
- The change in refractive index causes refraction rather than total internal reflection.
- When light passes through a layer with a lower refractive index the light will bend back to the fiber axis. Total internal

reflection does not occur because refraction fold the light back into the fibre axis. Before it reaches the cladding boundary.

- The light travels forward in the form of sinusoidal oscillation.
- The graded index profile which at present produces the best result. for multimode optical propagation. have a ^{near} parabolic refractive index profile with $\alpha=2$ fibre with such core index profile are well established and consequently when the curve graded index used without qualification it usually refers to a fibre with this profile.



$\alpha = 2$ $\therefore x$ = Radial distance from fiber axis

$$n(x) = \begin{cases} n_1 \left(1 - 2\Delta \left(\frac{x}{a} \right)^\alpha \right) & \text{when } x < a \text{ (core)} \\ n_1 (1 - 2\Delta)^{1/2} \approx n_2 & \text{when } x \geq a \text{ (cladding)} \end{cases}$$

⇒ LASER:-

→ The full form of LASER is Light Amplification by Stimulated Emissions of Radiation.

→ Laser is a device which produces a highly intense concentrated and parallel beam of monochromatic and coherent light.

→ Electromagnetic radiations is emitted whenever a charge particle such as an electron gives up energy.

→ Every time an electron drops from a higher energy state (E_1) into lower energy state (E_2) in an atom.

→ The colour of light is determined by a frequency or wavelength.

→ The shorter wavelength are ultraviolet and longer wavelength are infrared.

→ The smallest particle of light energy is described by quantum mechanics as a photon.

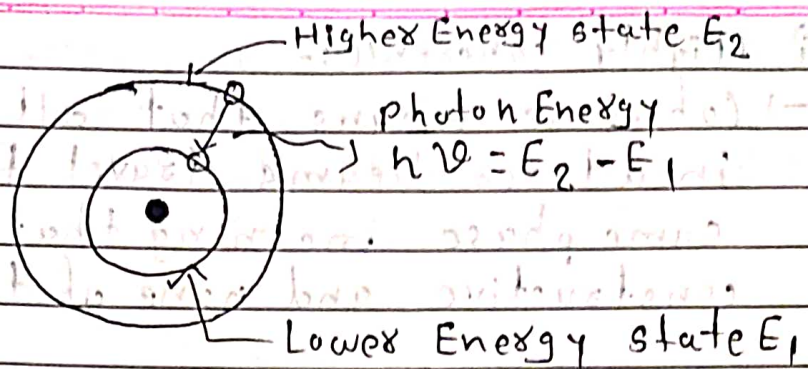
→ The energy (E) of a photon is determined by the frequency (ν) and Planck constant (h).

$$E = h\nu \quad \rightarrow \text{frequency}$$

→ The velocity of light in vacuum c is $3 \times 10^8 \text{ m/s}$.

→ The wavelength λ of light is related to ν from the following equation.

$$\therefore \lambda = \frac{c}{\nu}$$



→ Properties of Laser Light:-

1) Highly monochromatic:-

→ Laser Light is highly monochromatic.

Here monochromaticity refers to how pure in colour (frequency) or (wavelength) the laser beam is.

→ In other words how narrow the laser beam frequency bandwidth is.

2) Highly Directional:-

→ The Laser light is extremely directional or collimated light.

→ Collimated light is light in which or of the light rays are travelling in a specific direction and hence they are all parallel to each other.

3) Extremely intense or brightness:-

→ The brightness of laser arises from two factors. First of all the fact that the light is emitted in a well-defined beam means that the power per unit area is very high.

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④ Highly coherence:-

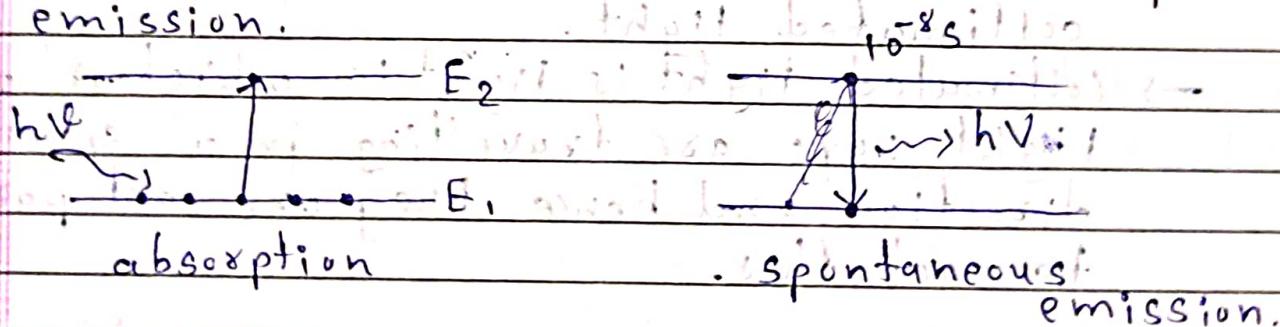
→ Coherence means that all the light waves in a laser beams travel together with the same phase meaning that they have constructive and none of the power wasted.

⑤ Focusability:-

→ many application of laser involves their ability to be focused to a very small spot size.

⇒ Interaction of process light with matter:-

→ There are three types of process involving the interaction of light beams with atoms that have electron recycling various energy levels for a two level atomic system and important for the laser. This are absorption, stimulated emission, and spontaneous emission.



→ Absorption:- When an incoming photon excites the atomic system from a lower energy level to a higher energy level is called absorption or stimulated absorption.

→ When a photon of energy $h\nu$ strikes an atom of the laser medium in the state

even and disappears exciting the atoms to the higher energy state E_2 .

→ The photon can only be absorbed if the absorption energy is $h\nu = E_2 - E_1$.

$$\nu = \frac{E_2 - E_1}{h}$$

E_1 → energy level -1 or ground level -1

E_2 → energy level -2 or excited level

h → planck constant.

→ Rate of stimulated absorption $R_{12}(\text{abs})$ - from level 1 to 2 is given as

$$R_{12}(\text{abs}) = B_{12} N_1 \cdot \nu(V)$$

→ where B_{12} is the Einstein coefficient for stimulated absorption and has the units as $\text{cm}^3/\text{s}^2\text{J}$.

→ Stimulated Spontaneous Emissions:-

→ Each level has a specific lifetime (τ) over which electron will remain in that level before decaying to a lower level.

→ Once the atom or molecule has been produced in its excited state there is a probability that it will emit radiation again and return to the lower energy level.

→ This lower energy state may be either the ground state or still one of the excited state but having lower energy.

→ In the process a photon is emitted.

→ The process in which atoms naturally goes to a lower energy state through the emission of

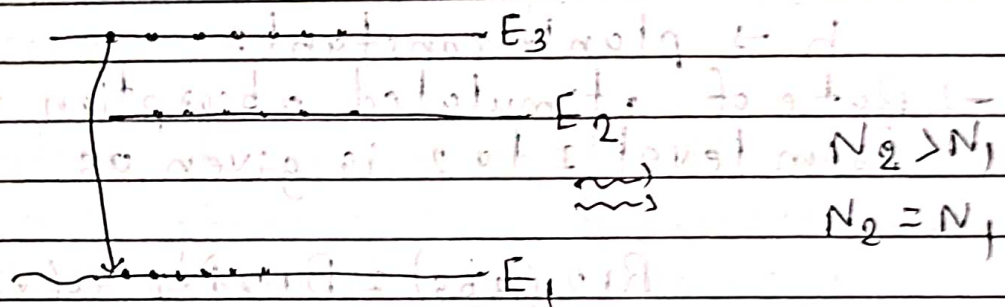
photon is known as spontaneous emission.

→ The rate of spontaneous emission

$$R_{21}(\text{spont}) = A_{21} N_2$$

→ where A_{21} is the Einstein coefficient of spontaneous emission and has the unit of $1/s$.

→ Stimulated Emission:- Induced Emission



→ It is also possible that the emission is stimulated by incoming photons which is called stimulated emission. It is when a photon of proper frequency interacts with a system of two energy levels. It is not essential that this photon will always be absorbed by ground state electrons and make a transition to excited state, as in case of absorption.

→ It may interact it to emit a fresh photon make a transition to ground state. This process is known as induced emission or stimulated emission. Both the photon incident and induced have the same phase direction spectral and polarisation properties and are in all aspects indistinguishable.

Relation between Einstein Coefficient:-

→ Under thermalequilibrium the population N_2 and N_1 of levels E_2 and E_1 respectively governed by the fact that the rate of upward transitions by absorption should be equal to the rate of downward transition by emission

→

$$\text{Absorption} = \text{Spontaneous Emission} + \text{Stimulated Emission}$$

$$B_{12} N_1 u(\nu) = A_{21} N_2 + B_{21} N_2 u(\nu) \quad \text{--- (1)}$$

$$A_{21} N_2 = B_{12} N_1 u(\nu) - B_{21} N_2 u(\nu)$$

$$A_{21} N_2 = u(\nu) [B_{12} N_1 - B_{21} N_2]$$

$$u(\nu) = \frac{A_{21} N_2}{[B_{12} N_1 - B_{21} N_2]} \quad \text{--- (2)}$$

divide numerator and denominator by $N_2 B_{21}$

$$u(\nu) = \frac{A_{21} N_2}{N_2 B_{21}} = \frac{\left(A_{21} / B_{21} \right)}{\left[\frac{B_{12} N_1}{N_2 B_{21}} - \frac{N_2 B_{21}}{N_2 B_{21}} \right]} \quad \text{--- (3)}$$

From Boltzmann's relation

$$\frac{N_1}{N_2} = e^{(E_2 - E_1 / kT)} \quad \text{--- (4)}$$

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Substitute the value of $\left(\frac{N_1}{N_2}\right)$ from eqn (4) to eqn (3).

$$u(\nu) = \frac{A_{21}}{B_{21}} \left[\frac{B_{12}}{B_{21}} e^{\frac{E_2 - E_1}{kT}} - 1 \right] \quad \text{--- (5)}$$

From plank's energy distribution formula

$$u(\nu) = \left(\frac{8\pi h}{\lambda^3} \right) \left[\frac{1}{e^{\frac{E_2 - E_1}{kT}} - 1} \right] \quad \text{--- (6)}$$

comparing eqn (5) & (6)

$$\frac{B_{12}}{B_{21}} = 1 \quad \text{or} \quad B_{12} = B_{21}$$

$$\text{Also } \frac{A_{21}}{B_{21}} = \frac{8\pi h}{\lambda^3} = \frac{8\pi h \nu^3}{c^3}$$

$$B_{21} = \frac{A_{21} \lambda^3}{8\pi h} = \frac{A_{21} \cdot c^3}{8\pi h \nu^3} = B_{12}$$

→ This relation conclude that:-

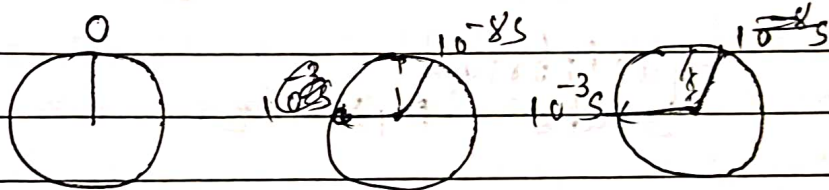
- ① Stimulated emission and absorption are competing processes.
- ② The rate of spontaneous emission is much larger than the induced emission.
- ③ The ratio A_{21}/B_{21} depends directly on the cube of the frequency of emission and inversely cube of wavelength.

Does in microwave region stimulated emission is probable then spontaneous emission and optical region spontaneous emission is more likely than stimulated emission.

Necessary Condition for laser action:-

1) Meta stable state:-

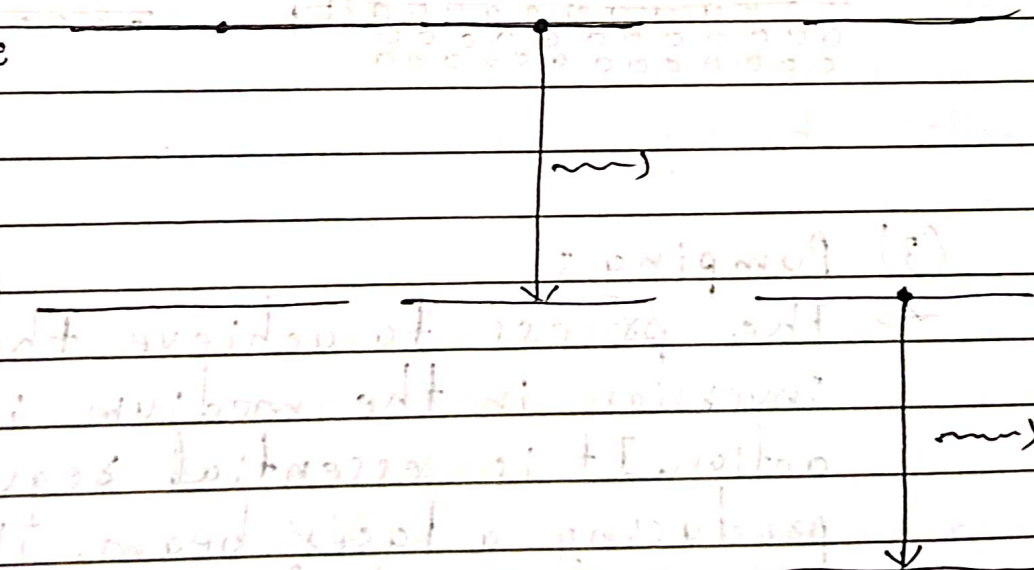
→ The key to the laser is the presence of many atoms of one or more excited level whose life times may be 10^{-3} s or more. ~~More~~ instead of the usual 10^{-8} s. Such relatively long live states are called meta stable state.



ordinary
excited state

meta stable
state

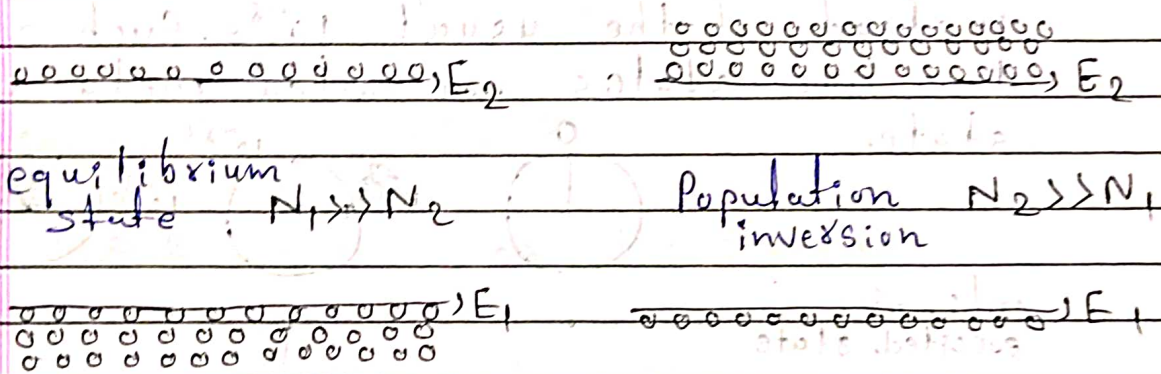
Ground
state



(2) Population inversion :-

→ Population inversion creates a situation in which the no. of atoms in higher energy state is more than that in the lower energy state.

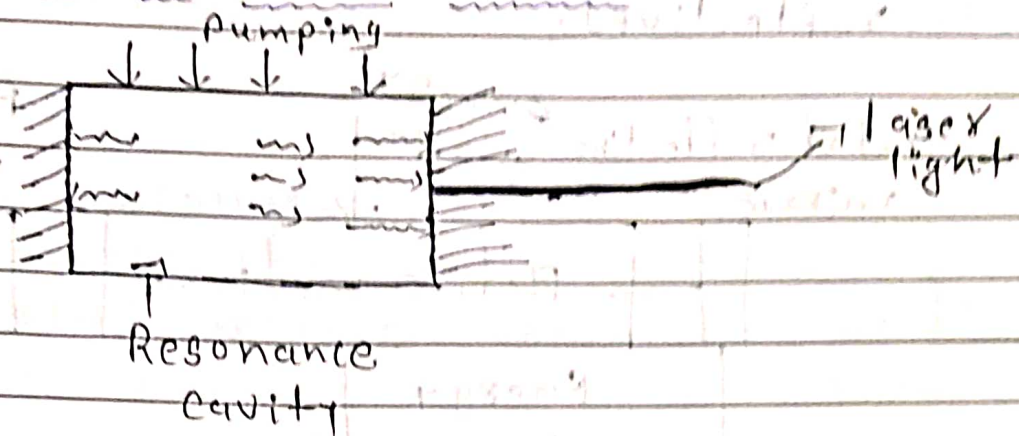
→ Usually at thermal equilibrium the no. of atoms N_2 that is population of atoms at excited state is much lesser than the population of atoms at ground state N_1 .



(3) Pumping :-

→ The process to achieve the population inversion in the medium is called pumping action. It is essential requirement for producing a laser beam. The methods commonly used for pumping are ① Optical pumping ② electrical discharge ③ Direct conversion, ④ In elastic atom-atom collision between atoms.

Component of laser device:-



1) Active medium:-

- It is the material in which the laser action takes place. The active medium may be solid crystals such as ruby, liquid dyes, gases like CO_2 or Helium/Neon, such as GaAs. This medium decides the wavelength of laser radiation. Active mediums contain atoms which can produce more stimulated emission than spontaneous emission and cause amplification they are called "Active Centers".

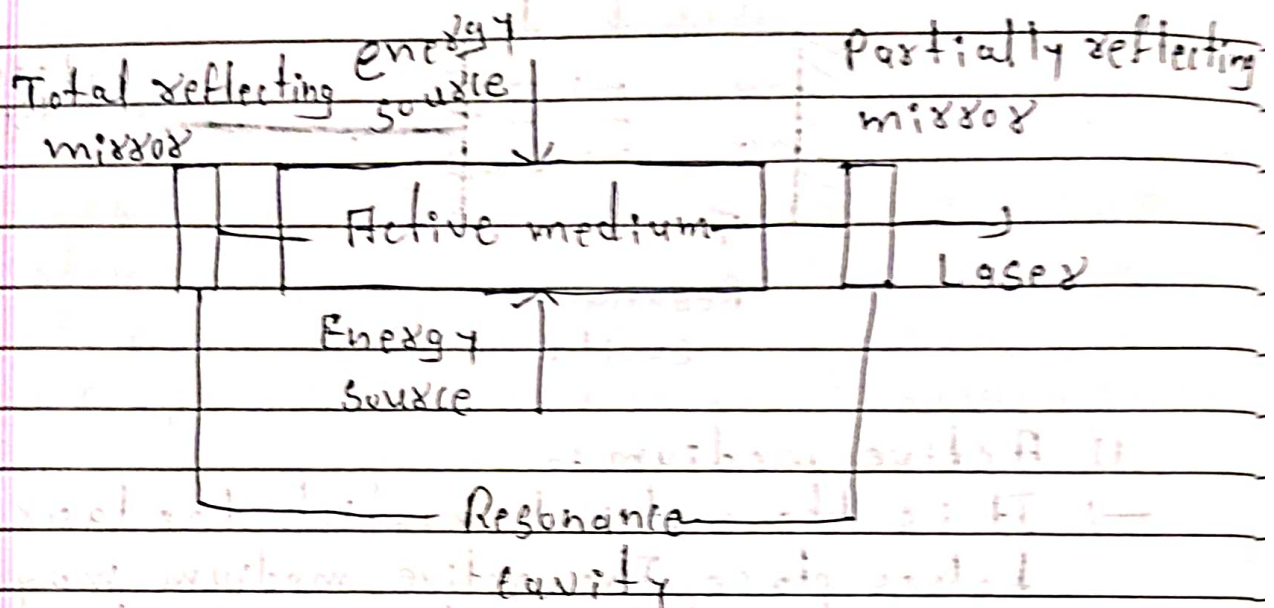
2) Pumping Energy Source (Excitation Mechanism):-

- Energy source pumps the active centers from ground state to excited state to achieve population inversion. The pumping by energy source can be optical, electrical or chemical depending on the active medium.

3) Resonance Cavity:-

- Resonance cavity of active medium enclosed by two mirrors one is highly reflective mirror (100% reflective) and the other is

is partially transmissive mirror (99% reflective).



Type:- Helium-Neon LASER:-

→ It is a gas laser developed by Ali Javan and his coworkers first introduced in 1961. It is developed because ruby has a disadvantage that it releases laser light in form of pulses and not in continuous form.

→ Construction:-

① Active medium:-

→ He-Ne laser consist of a narrow tube of glass or quartz of length 10 to 100 cm with a diameter of 5 mm to 10 mm. The tube is filled with a mixture of He and Ne gases approximately in the ratio 85% He and 15% Ne at low pressure. Ne acts as at two centres atom

and responsible for laser action. While He atoms are used to help in the excitation process.

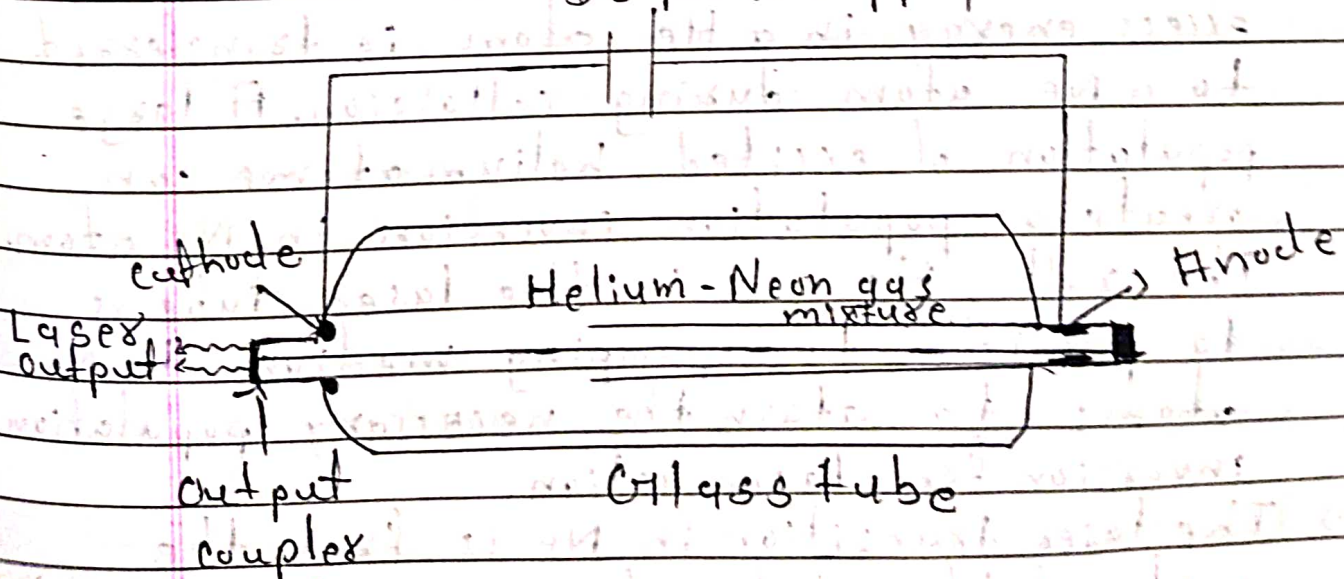
② optical resonator:-

- To construct the optical resonator cavity two parallel mirrors are placed at the ends of the quartz tube one of them partial transparent while the other is fully reflecting.
- The spacing betⁿ the mirror is adjusted such that it should be equal to the integral multiple of half wavelength of the laser light.

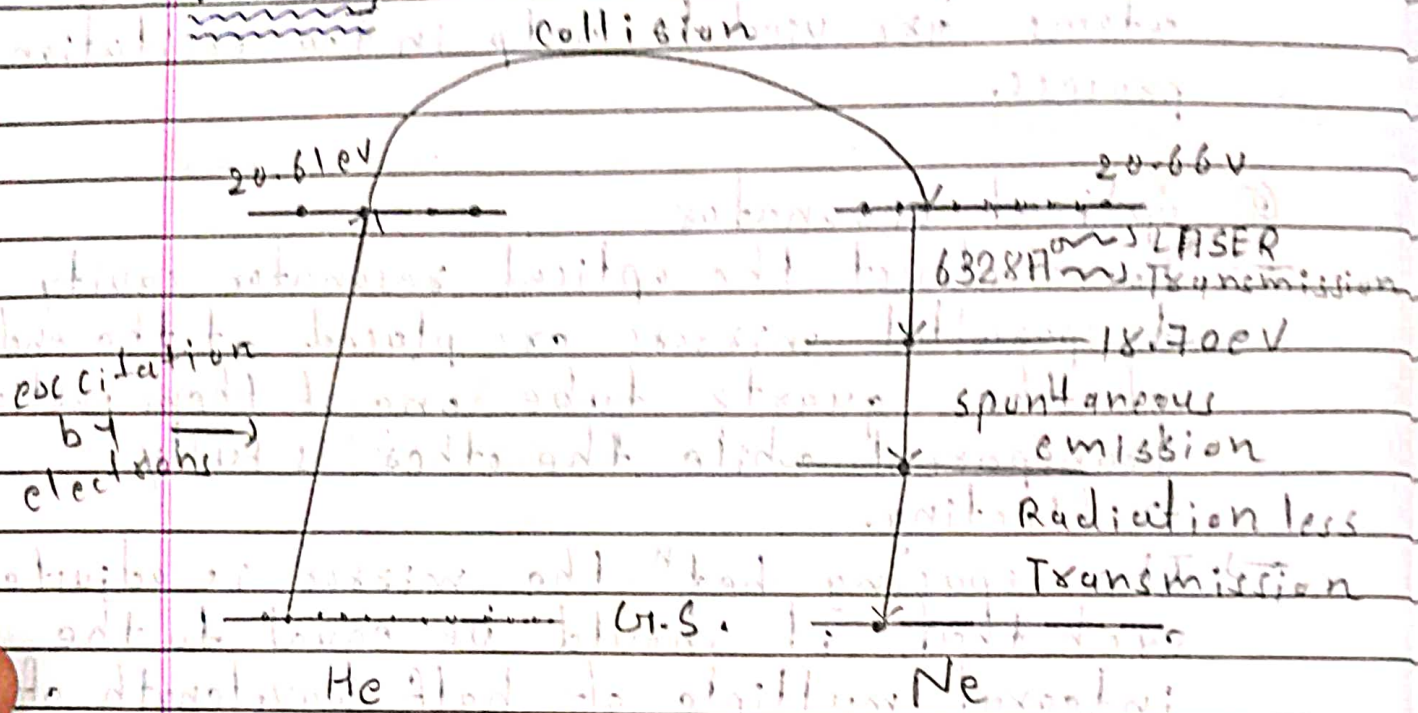
③ Pumping system:-

- The pumping is done through electrical discharge by using electrodes that are connected to a high frequency alternating current source.

DC power supply



→ Working:-



→ The laser process in He-Ne laser starts with collision of electrons with the helium atoms.

→ In the gas by electrical discharge. This electrical discharge excites helium atoms from the ground state to the long lived meta stable excited state. The excess energy in a He atom is transferred to a Ne atom during collision. A large population of excited helium atoms can create a population inversion in Ne atom. The role He gas in the laser tube is to provide the pumping medium to Ne atoms to attain the necessary population inversion for laser action.

→ The laser transition in Ne is from the meta stable state at 20.66 eV to an excited state at 18.70 eV with the emission of a

632.8 $\text{\AA}$ 632.8 $\text{\AA}$ photons.

→ Then another photon is spontaneously emitted in transition to a lower meta stable state. This transition emits only in coherent light. The remaining exciting energy is lost in collision with the tube walls.

→ This laser emits red beam.

* The electron that excites the He-Ne atoms occurs all the time unlike the pulsed excitation from the xenon flashed lamp in a ruby laser.
A He-Ne laser operates continuously.

Semiconductor Diode Lasers:-

→ It is specifically fabricated p-n junction diode. This diode emits laser light when it is forward biased.

→ Characteristics:-

→ It is a solid state semiconductor laser.

→ A PN junction diode made from single crystal of gallium arsenide is used as an active medium.

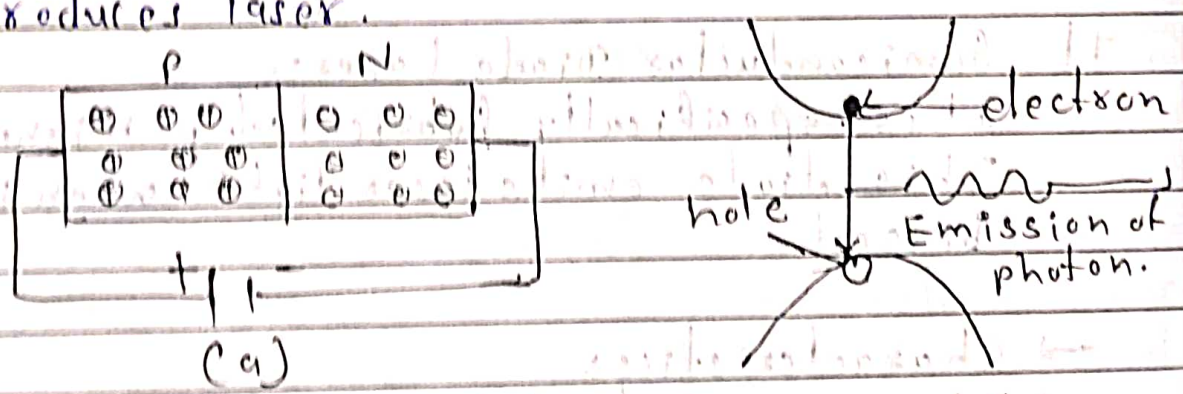
→ The direct conversion method is used for pumping action.

→ The power output from this laser is 1 mW.

→ The nature of o/p is continuous wave or pulsed o/p.

→ Gallium arsenide laser gives infrared radiation in the wavelength 8300 to 8500 $\text{\AA}$.

- Principles:
- When a p-n junction diode is forward biased, the electrons from n-region and the holes from the p-region cross the junction and recombine with each other.
 - During the recombination process, the light radiation is released from a certain specified direct band gap semiconductor like Ga-As. This light radiation is known as recombination radiation. The photon emitted during recombination stimulates other electron and holes to recombine. As a result, stimulated emission takes place which produces laser.



→ Working: In Pdf. (Advantage, disadvantage, Application)

→ Construction:-

→ Figure shows the basic construction of semiconductor laser. The active medium is a p-n junction diode made from the single crystal of gallium arsenide. This crystal is cut in the form of a platter having thickness of 0.5 mm .

→ The platter consists of two parts having an electron conductivity (n-type) and hole conductivity (p-type). The photon emission is stimulated in a very thin layer of PN junction. The electrical voltage is applied to the crystal through the electrode fixed on the upper surface. The end faces of the junction diode are well polished and parallel to each other. They act as an optical resonator through which the emitted light comes out.